A Returns-Only Audit of Market Fragility and Early Warning

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Version 1.1 – 6 November 2026

Abstract

I evaluate a fixed, returns-only audit of market-state summaries and a local susceptibility feature. Participation entropy stays near ten, so I find no strongly concentrated low-dimensional mode under this measure. Rotation is weak and near-degenerate, and Wedin's bound is informative on only 2.53% of test dates. In the frozen analysis, F is the only signal to survive the block-shuffle surrogate test (Holm-adjusted $p = 0.006$). A post-freeze 4,999-shuffle rerun gives adjusted $p = 0.0036$ under the frozen settings, but $p = 0.0804$ with a 10^{-6} floor and $p = 0.0306$ when exact-zero observations are treated as missing; the pre- and post-2000 adjusted values are 0.087 and 0.0732. Thus, F survives under the frozen settings, but not under every alternative zero-handling or subperiod check. The observed median F is below the shuffled null, indicating less amplification than in shuffled data. Adding F to the richer forecasting baseline changes AUC by -0.009, with an adjusted upper interval bound of 0.004

predeclared comparison structure to reduce post hoc selection, while recognizing that a plan cannot remove modeling choices or low event counts [12]. I measure local response in filtered returns; I do not identify a structural crisis mechanism.

I make four contributions:

- I give a reproducible, returns-only audit that separates descriptive state measures from out-of-sample forecast value.
- I report signal-level surrogate and pre-onset results alongside the frozen joint H1 decision.
- I test whether local fragility adds AUC beyond two matched baselines and report the post-freeze sensitivity analyses separately.
- I release the implementation, frozen derived results, and analysis addendum with explicit data and file hashes.

I first describe the data, filter, local response map, and decision rules; I then report validation and frozen results, followed by the exploratory addendum, interpretation, limitations, and conclusion.

2 Method

2.1 Data and filtering

I use daily value-weighted industry returns from the French Data Library. I convert each simple return $u_{i,t}$ to the continuously compounded value $r_{i,t} = \log(1 + u_{i,t})$. The source uses designated missing-value codes; I exclude a date if any portfolio has a coded missing value. An exact zero simple return remains zero in the frozen analysis; its variance measurement is $\log c = -18.42$ when $c = 10^{-8}$. I count 442 exact-zero cells among 69,430 portfolio-date cells in the 1926–1949 estimation window.

I filter each industry's log variance $h_{i,t}$ with a diagonal Gaussian quasi-likelihood Kalman filter:

$$h_{i,t} = \mu_i + \phi_i(h_{i,t-1} - \mu_i) + \eta_{i,t}, \qquad y_{i,t} = \log(r_{i,t}^2 + c) = h_{i,t} + m + \xi_{i,t}. \tag{1}$$

Here $c = 1e-08$, $m = -1.27$, and the variances of η and ξ are q_i and $R_\xi = 4.935$. I use the known log-chi-square mean and Gaussian observation noise as a quasi-likelihood approximation [2]. For the linear transition and observation, the unscented transform in the original prototype gives the same moments as the Kalman recursions; I use the direct recursions in C++ and verified their numerical equivalence [1].

I estimate parameters only in the estimation window. For each date, I standardize filtered states using an expanding past-only mean and standard deviation, with floor 0.05 and clipping at plus or minus 6. The bounds, preprocessing rules, and date splits remain those in the frozen protocol.

2.2 Local susceptibility and state summaries

I fit an online recursive least-squares map with intercept and forgetting factor $\lambda = 0.998$:

$$\mathbf{z}_{t+1} = g_t(\mathbf{z}_t), \qquad J_t = \mathrm{D}g_t(\mathbf{z}_t) = \frac{\partial g_t(\mathbf{z}_t)}{\partial \mathbf{z}_t}. \tag{2}$$

The Jacobian J_t is the local derivative of the fitted map. For a small perturbation $\delta\mathbf{z}_t$, I use $\delta\mathbf{z}_{t+1} \approx J_t\delta\mathbf{z}_t$ as a first-order description: it asks how a small change in today's estimated market

state would change the next fitted state. I define fragility at horizon 5 as $F_t = \|J_t^5\|_2$. It measures the largest local amplification over that horizon. It is a statistical sensitivity measure, not a causal effect or a guarantee of future market behavior.

I define rotation R_t as the mean principal angle between leading right-singular subspaces of J_t separated by 20 sessions. I define participation entropy as $H_t = \exp[-\sum_i p_{i,t} \log p_{i,t}]$, where $p_{i,t} = s_{i,t}^2 / \sum_j s_{j,t}^2$. This effective mode count is invariant to rescaling J_t and lies between one and the state dimension [7]. A value near ten indicates that singular energy is not concentrated in a small number of modes; it does not establish that collective behavior is impossible.

I summarize stress with annualized realized volatility, drawdown from the trailing peak, and past-relative average pairwise correlation. I also use an AR(1) coefficient and absorption ratio as established comparators [9, 8]. Funding, leverage, and liquidity are unavailable in this returns-only panel. For singular subspaces of the non-symmetric Jacobian I use Wedin's perturbation bound; I use Davis–Kahan for symmetric covariance matrices [4, 3].

2.3 Forecast target and inference

I define an event as the first crossing of a drawdown of -15% from the trailing peak. Crossings less than 120 sessions apart form one cluster. A positive label marks an onset within the next 60 sessions, and I remove dates inside event clusters from the risk set. I compare $M_0 = \mathbf{S}$, $M_0 + F$, $M_1 = (\mathbf{S}, \text{AR}(1), \text{absorption ratio})$, and $M_1 + F$. I fit expanding logistic models with $C = 1$, refit every 252 sessions, and purge the final 60 training dates.

I call an anchor an out-of-sample risk date with an observed target and a prediction for every model in the comparison. My primary incremental statistic is $\Delta\text{AUC}_1 = \text{AUC}(M_1+F) - \text{AUC}(M_1)$. I use circular blocks of 250 sessions, about one trading year and longer than the 120-session event-cluster window, to retain low-frequency dependence in score-label pairs [5]. This is a declared bandwidth rather than an optimized one; I compare 60-, 125-, and 250-session blocks in a separately planned post-freeze check. I adjust the two primary model comparisons with Holm's sequentially rejective procedure [6].

I call H2 supported only when its adjusted lower interval is above zero and its point increment reaches 0.02. I call it unsupported when the adjusted upper interval is below that practical threshold; otherwise I call it inconclusive. Thus, an interval that includes zero can still rule out the prespecified practical improvement.

2.4 Frozen H1 decision rule

I evaluate four conditions together. Condition (a) requires all three Holm-adjusted block-shuffle probabilities for R, H, and F to be at most 0.01. Condition (b) requires at least one Holm-adjusted pre-onset placebo probability below 0.05. Condition (c), redundancy, requires the maximum absolute Spearman correlation between F and the observed stress features to be below 0.7. Condition (d), dimension, requires at least three principal components to explain 90% of the standardized state vector's variance. I require all three signals to pass condition (a) because H1 concerns the full vector as a set of distinct state summaries; evidence for one component alone cannot establish that joint claim. I support H1 only if all four conditions pass. The protocol's unsupported rule is "all adjusted block-shuffle probabilities are at least the pre-onset placebo threshold." I apply that rule literally: the joint H1 result is unsupported only when every adjusted block-shuffle probability is at least the pre-onset placebo threshold. I report per-signal findings even when the joint rule is inconclusive.

3 Validation and frozen results

3.1 Implementation checks

I passed the declared numerical, estimator-recovery, synthetic-control, filtering, and pre-test checks before the protocol freeze. I summarize the gates in Table 1; the supplement gives the full definitions and measured quantities. I retain the original frozen study output; later analyses are recorded separately.

Table 1: Validation gates. I report measured values against thresholds fixed before the primary analysis.

Gate	Value	Threshold	Result
U	UKF error 1.2e-14; finite difference 9.8e-11	$\leq 1 \times 10^{-10}$; $\leq 1 \times 10^{-9}$	pass
E1	10/10 assets	$\geq 8/10$ and no box boundary	pass
G1	error 0.03818; correlation 0.8608	≤ 0.35; ≥ 0.5	pass
G2	rotation 0.2618; log amplification 2.312	≥ 0.15; ≥ 0.15	pass
G3	RMSE improvement 59.27%	$\geq 5\%$	pass
G4	median rho 0.988; median F5 1.071; max F5 1.329; direct gap 0.08337	≤ 1.05; ≤ 3; ≤ 1000; ≤ 0.5	pass
Attempts	5	at most 5	pass

3.2 Directions and state summaries

I compare the leading covariance eigenspace with the leading susceptibility right-singular subspace. The observed directions are closer than most Haar reference subspaces: the median date-wise percentile is 5.9% across 14308 dates. This describes alignment, not causation. Wedin's bound is informative on 2.5% of dates.

I account for serial dependence in the date-wise Haar percentiles with a 250-session circular-block bootstrap. I obtain median 5.9% (95% block interval [3.5%, 8.2%]); effective sample size 49.69 of 14308 dates; the supplement also reports the factor-aligned null.

I report six distinct summaries rather than combining them into one score. Figure 2 shows the test-period paths. H stays close to the ten-dimensional ceiling and changes little; R varies but has no stable surrogate-exceeding evidence. In this analysis, H is near-constant and R is noise-dominated, which undercuts H1's premise that these summaries provide a useful set of distinct state signals. This pattern is consistent with a near-identity, highly persistent fitted J (median $\rho \approx 0.99$), whose singular values are close.

3.3 H1 signal and joint outcomes

I find that only F passes the adjusted block-shuffle test: $p_R = 0.844$, $p_H = 0.984$, and $p_F = 0.006$. For the pre-onset placebo, I report raw and Holm-adjusted probabilities side by side: R is $(0.3846, 1)$, H is $(0.8142, 1)$, and F is $(0.8252, 1)$. Holm adjustment caps all three large raw values at one; I do not interpret these adjusted values as evidence that the observed statistics equal their placebo medians.

I find H1 inconclusive. Condition (a) fails jointly because R and H do not meet the 0.01 surrogate threshold, although F passes individually. Condition (b) fails because none of the adjusted

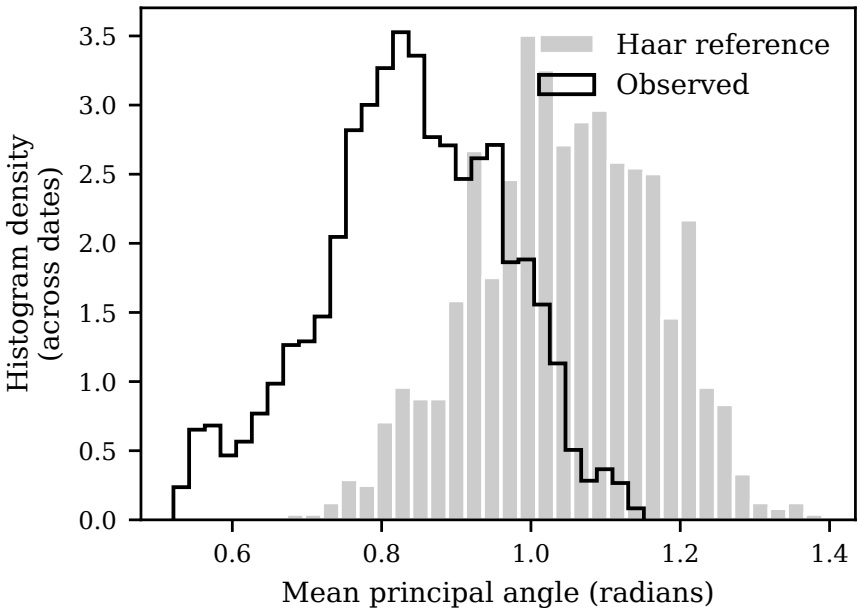


Figure 1: Histogram density across test dates for the mean principal angle. I show the Haar reference distribution and observed covariance-to-susceptibility angles ($n = 14308$ dates).

pre-onset probabilities is below 0.05. Conditions (c) redundancy and (d) dimension pass. The frozen unsupported rule therefore does not apply, so the joint verdict remains inconclusive.

Table 2 gives the four frozen condition values and rules.

Table 2: Frozen H1 conditions and measured values. The pre-onset entries show raw/Holm-adjusted probabilities.

Condition	Frozen result	Rule	Outcome
(a) Surrogates	$p_R = 0.844$, $p_H = 0.984$, $p_F = 0.006$	$p \leq 0.01$ for all	fail
(b) Pre-onset	raw/Holm: R: 0.385/1; H: 0.814/1; F: 0.825/1	adjusted $p < 0.05$ for one or more	fail
(c) Redundancy	$0.4793 < 0.7$	maximum absolute Spearman correlation	pass
(d) Dimension	4 components	≥ 3 for 90%	pass

3.4 H2 incremental prediction

I find H2 unsupported. Table 3 reports both frozen model comparisons. For the richer baseline, adding F changes AUC by -0.008816, with adjusted interval $[-0.02427, 0.004163]$. Its point estimate is negative, and the adjusted upper bound is below the practical threshold of 0.02. The test contains 21 event clusters and 10793 common anchors.

In the post-freeze default forecast comparison, EWMA volatility alone reaches AUC 0.748, at least as high as M_1 (0.741) and $M_1 + F$ (0.732). In Table S4, at the -10% threshold and 60- and 120-session horizons, AUC falls below 0.5 for M_1, $M_1 + F$, and EWMA. This may reflect the risk set: some high-volatility dates after an event can be labelled negative when no new onset falls within the forecast horizon. I treat this as a possible explanation, not a demonstrated cause.

I see several plausible reasons why adding F could lower AUC: it may overfit incremental variation when severe events are few; it may be collinear with stress already represented in M_1;

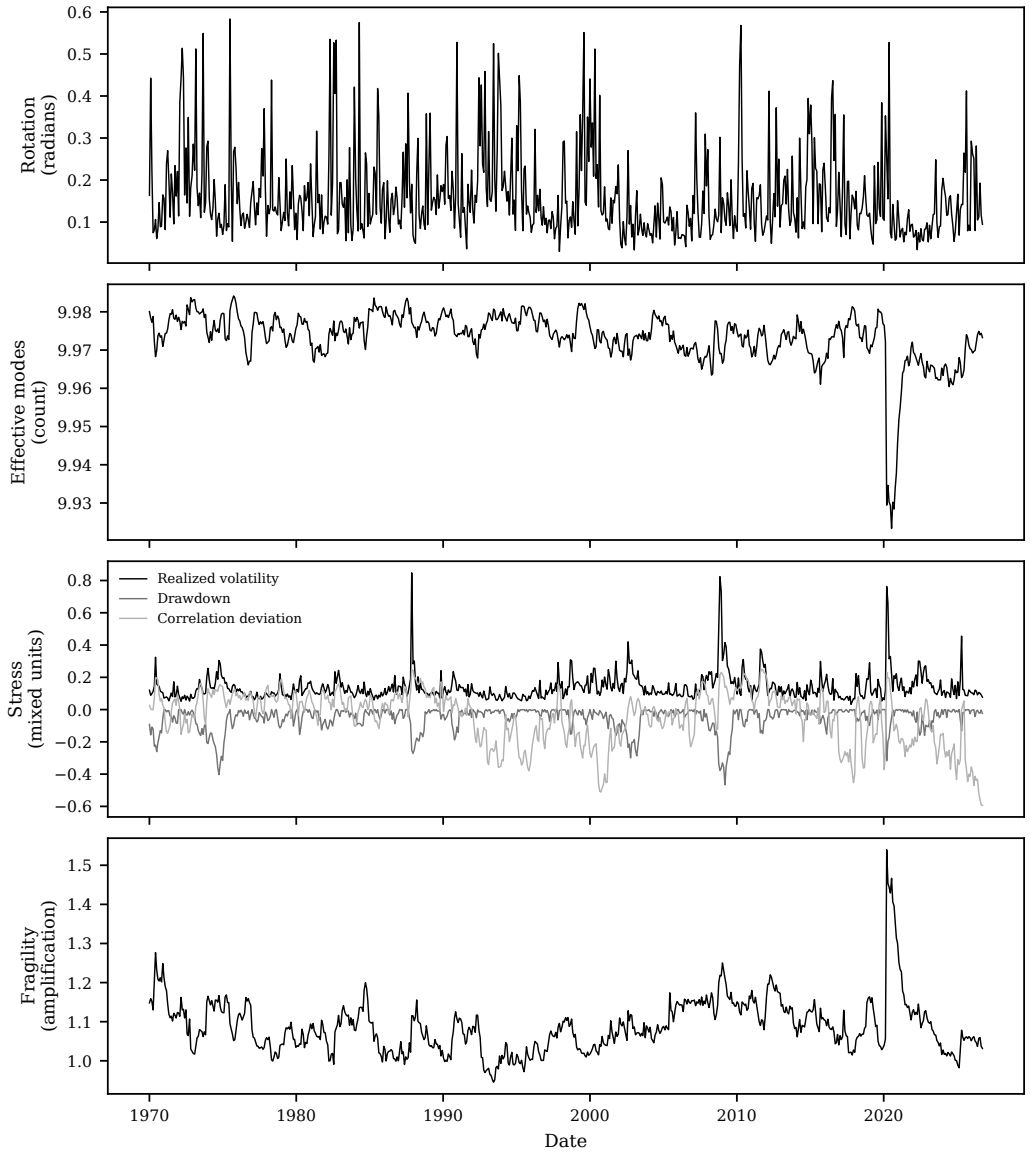


Figure 2: The six components of $\mathbf{M}_t$ over the out-of-sample period. Volatility is annualized; drawdown and correlation deviation are unitless.

Table 3: Frozen incremental out-of-sample AUC. An anchor is a risk date with an observed label and a prediction for each model in its comparison; I use the common anchors for each pair.

Comparison	ΔAUC	Adjusted interval	Holm p	Clusters	Anchors
M0+F vs M0	-0.01366	[-0.02799, 0.001076]	0.088	21	10793
M1+F vs M1	-0.008816	[-0.02427, 0.004163]	0.17	21	10793

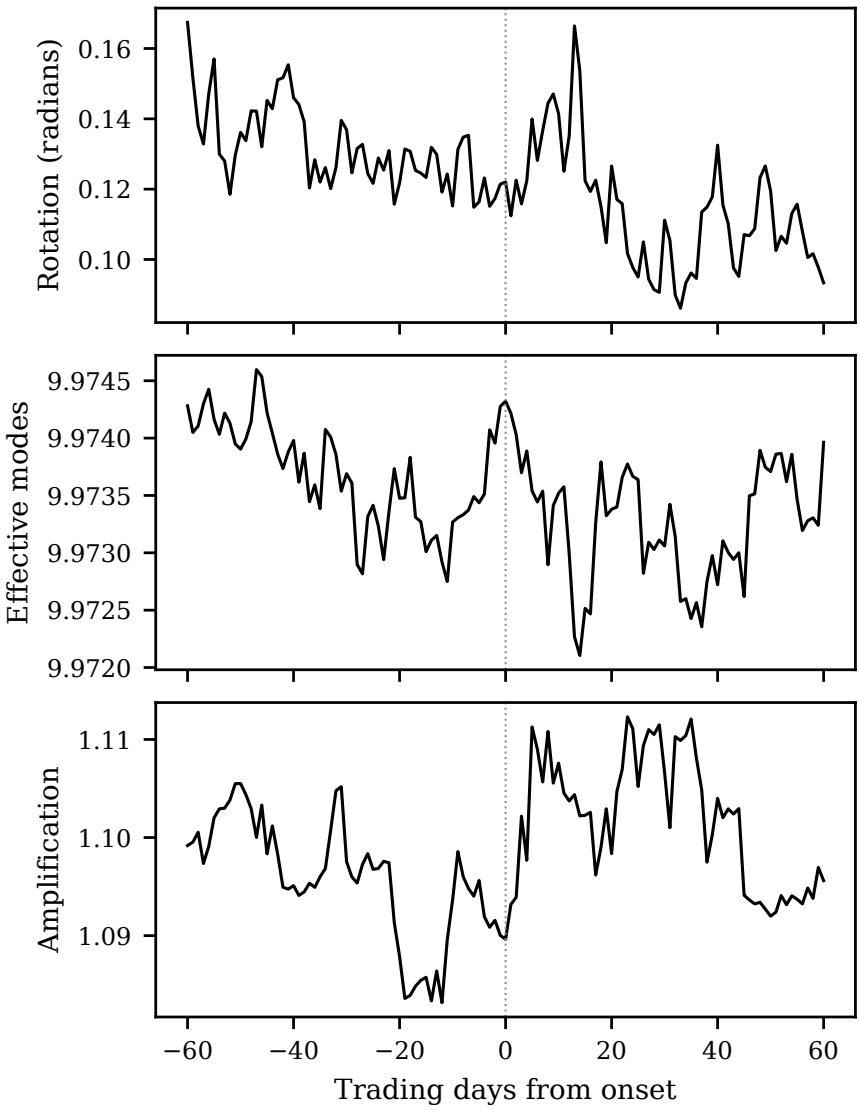


Figure 3: Median rotation, entropy, and fragility around each test-period event onset. I use the date alignment descriptively; it does not establish causality.

or it may move too slowly to help a model fit on a small number of event clusters. These are explanations to test, not conclusions established by this comparison.

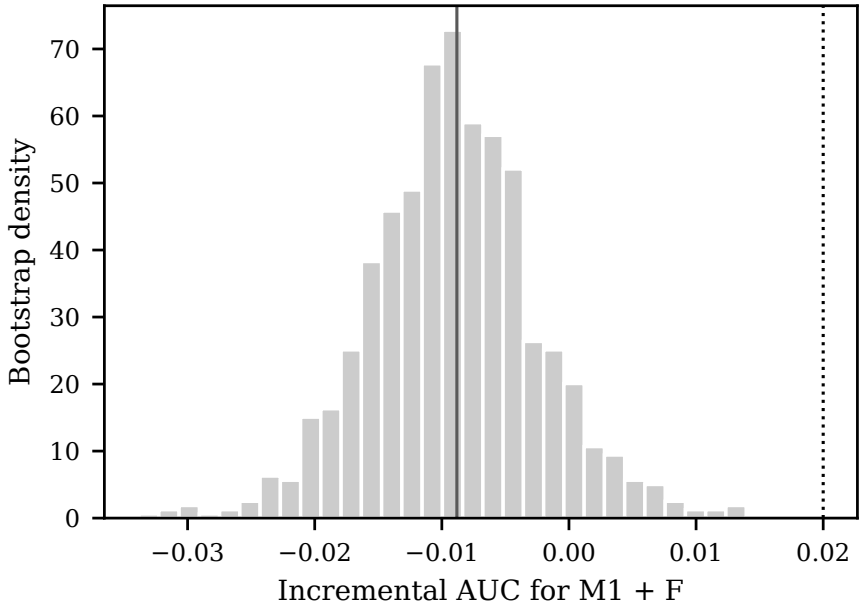


Figure 4: Frozen block-bootstrap distribution of the richer-baseline AUC increment. I mark the point estimate and practical threshold.

4 Post-freeze exploratory addendum

After the protocol freeze, I wrote a separately hashed exploratory analysis plan and three separately hashed extensions, and I recorded each in the amendment log. I leave the primary analysis plan, protocol hash, and frozen study output unchanged. I use these additional analyses to assess sensitivity, not to revise the primary verdicts.

I count exact zero returns by estimation, calibration, and test period. I compare the frozen 10^{-8} floor with a 10^{-6} floor and with a filter treatment that skips the measurement update when a return is exactly zero. I keep zero source returns in the equal-weight target and stress series. I summarize the period counts and both reruns in the supplement.

For F, I rerun the primary, zero-treatment, subperiod, and forgetting-factor block-shuffle tests with 4,999 draws per test. In the primary rerun, the observed median is 1.08 versus a shuffled-null median of 1.12 (raw $p = 0.0012$, Holm-adjusted $p = 0.0036$). The floor-10^{-6} result is 1.16 versus 1.19 (raw $p = 0.0268$, Holm-adjusted $p = 0.0804$); with exact-zero observations treated as missing it is 1.09 versus 1.12 (raw $p = 0.0102$, Holm-adjusted $p = 0.0306$). The F surrogate result therefore depends on zero treatment: under the frozen 10^{-8} floor, $\log(0 + c) = -18.42$ turns each exact zero into an extreme outlier. In both subperiods, the observed median is below the shuffled null: 1970–1999: 1.06 vs 1.09 (raw/Holm $p = 0.087/0.087$); 2000–2026-09: 1.10 vs 1.14 (raw/Holm $p = 0.0366/0.0732$). The direction is lower amplification than shuffled data, not higher. These reruns show that F survives under the frozen settings, but not under every alternative zero-handling or subperiod check.

I supplement the Haar comparison with a factor-aligned null that fixes the leading covariance direction and randomizes the remaining directions in its orthogonal complement. I report block-bootstrap uncertainty and an autocorrelation-adjusted effective sample size for both the frozen Haar percentile and the factor-aligned percentile. The factor-aligned result is less extreme: I obtain 18.1%, compared with 5.9% for the frozen Haar reference. This suggests that part of the apparent alignment reflects the market's leading factor. Across $\lambda \in \{0.990, 0.995, 0.998, 0.999\}$, the post-freeze Holm-adjusted F probabilities are 0.0024, 0.0078, 0.003, 0.0006. The median scale of F depends on λ: it ranges from 1.03 at 0.999 to 1.36 at 0.990, so the selected-setting median of 1.08 is not an absolute level. I find the entropy ceiling robust, but the scale of rotation sensitive to the forgetting factor.

For H2, I vary the drawdown threshold over -10%, -15%, and -20%, the forecast horizon over 20, 60, and 120 sessions, and logistic C over 0.1, 1, and 10. I report AUC levels and increments against a no-skill AUC of 0.5 and an EWMA-volatility-only baseline, along with precision and recall at a 5% alert rate. A separate 399-replicate bootstrap resamples training blocks and refits the logistic models at the original expanding-window cutoffs; I hold the upstream filter and RLS path fixed. I also compare fixed-score interval bandwidths of 60, 125, and 250 sessions.

Across the 27 H2 settings, I estimate a positive AUC increment in six cases, but no point estimate reaches the 0.02 practical threshold. The upper interval remains below 0.02 in 20 settings; seven settings leave that threshold inside the interval. I report those less decisive cases rather than treating the frozen negative result as universal across target definitions.

For the frozen target and fitted scores, the fixed-score interval widens as the block length increases: I estimate intervals of $[-0.019, 0.002]$, $[-0.020, 0.003]$, and $[-0.021, 0.005]$ for 60, 125, and 250 sessions. Each upper endpoint remains below 0.02.

I repeat the analysis on the 17-industry daily panel as an aggregation check. I match 19 of its 22 test clusters to ten-industry events within 120 sessions (86.4%). Its raw F surrogate probability is 0.002 from 499 draws, while the primary F probability is Holm-adjusted after the 4,999-draw rerun; these values are not comparable. Adding F changes H2 AUC by -0.024 (95% interval $[-0.052, -0.001]$), which excludes zero and indicates that F hurts on this panel, across 10,288 anchors. The heavy event overlap adds little independent H2 event power, so I do not treat this as a replication. I found no daily developed ex-U.S. industry series in the official French Library industry tables; its daily developed ex-U.S. portfolios are sorted by size and book-to-market, so I do not present them as an industry panel [13]. I report the exploratory results and their hashes in the supplement.

5 Discussion

I draw a narrow conclusion from the frozen analysis: F survives the block-shuffle null, but it does not add practical forecast value under H2. Entropy is near-constant and rotation is noise-dominated; both findings undercut H1's premise. F survives under the frozen settings, but not under every alternative zero-handling or subperiod check. The post-freeze checks show how the result responds to zero treatment, null construction, event definition, model penalty, time period, forgetting factor, and portfolio aggregation; none alters the original joint verdict.

I interpret F as a local sensitivity of a fitted response map. In future work, I could model a probability density over market states with a nonlinear drift and state-dependent diffusion, then use the Jacobian of the drift to describe local susceptibility. A nonlinear Fokker–Planck equation is one possible framework. I propose this only as a direction for later research; I do not estimate such a PDE here.

6 Limitations

- I use one U.S. historical return source and only 21 frozen test-period event clusters. The 17-industry check reaggregates largely the same stock universe and does not supply independent international event power.
- My stress features omit funding, leverage, and liquidity. I forecast historical drawdowns and do not estimate a traded strategy or net performance.
- My refit-aware bootstrap resamples logistic training and test blocks but holds the filtered state and RLS path fixed, so it does not include all upstream estimation uncertainty.
- I disclose two access deviations: a final-row preview and pre-test data in an early lambda-selection harness. I saw no test-period aggregate or model result; I give the chronology in Supplementary Section 11.
- The French Library can revise historical return files. I record snapshot hashes and provide a downloader, but an updated archive may not reproduce the frozen snapshot.

7 Conclusion

I find H1 inconclusive and H2 unsupported under the frozen decision rules. F survives under the frozen settings, but not under every alternative zero-handling or subperiod check; adding it to the richer baseline does not achieve the prespecified practical AUC gain. Entropy stays near its ten-mode ceiling and rotation does not show robust surrogate evidence. The observed median F is below the shuffled null, indicating less amplification than shuffled data. This audit therefore finds a setting-dependent local-sensitivity signal without evidence that it improves the selected drawdown forecasts.

Declarations

Funding. I received no grant or external funding, and I did not seek funding for this project.
Competing interests. I disclose that I first designed Avarent as a proprietary algorithm product for quantitative funds and later decommissioned the Avarent project.
AI assistance. AI tools substantially assisted me with code, analysis scripts, and drafting. I developed the initial framework on paper and the C++ prototype, reviewed the code, analyses, and manuscript, and take responsibility for the reported methods and results.
Data and code availability. I obtained the ten-industry data from the Kenneth R. French Data Library (archive retrieved 2026-11-05) and the 17-industry daily panel from the same Library (retrieved 2026-11-06). I provide scripts that fetch both archives from the repository root. I do not redistribute the upstream French archives; their source terms apply. My code is available at `https://github.com/FFTryhard/Avarent`. I license code under MIT and my paper, figures, and derived results under CC BY 4.0. I list SHA-256 hashes for the source archives, derived panels, frozen study file, analysis plans, protocol, and executable in the supplement's reproducibility section.
Disclaimer. I do not provide investment advice and make no claim of trading performance.
Acknowledgments and ethics. I acknowledge the Kenneth R. French Data Library. I use public portfolio-level market data and no human-subject or individual-level records.

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